

SKIIN Deep Research and Gap Analysis (Additional Findings)

This document captures additional research gathered after the initial copy specification, and highlights specific gaps and misstatements that need addressing in the final website copy.

1 SKIIN Technology & Electrode Origins

1.1 Textile Computing & Sensor Integration

SKIIN is built on **textile computing** – the integration of biosensing technology directly into fabric through knitting and conductive yarns. Sensors are sewn or knitted into garments, allowing continuous monitoring without external wires or adhesive patches. Myant Corporation's acquisition of Swiss company **Nanoleq** illustrates how these capabilities are being advanced. Nanoleq specialises in smart textiles and biosensing technologies; its **stretchable electrical cables and dry electrodes** provide the underlying sensing capability for devices like SKIIN. Myant has stated that **Nanoleq's advanced sensor capabilities will be integrated into its medical devices** ¹. Nanoleq's sensors can monitor respiratory function, ECG, HRV and stress levels, and are incorporated into smart garments ².

These findings confirm that SKIIN's sensors are **developed by Myant using proprietary textile computing techniques**, incorporating **Nanoleq's patented dry electrodes and stretchable conductive cables** ¹. Rather than affixing sensors onto fabric, Myant **weaves them directly into the garment through knitting**, embedding conductive yarns and electrodes into the textile structure. This enables comfortable, washable and unobtrusive monitoring and means the hardware is invisible to the wearer. There is **no evidence** that these sensors were engineered by ETH Zurich; accordingly, any copy claiming SKIIN's sensors were "Swiss-engineered in collaboration with ETH Zurich" is unsubstantiated and must be removed. Instead, accurately credit Myant's textile-computing platform and Nanoleq's contributions, emphasising the **knitted integration** of the monitoring capabilities.

1.2 Carbon Electrodes & Graphene Coatings

Previously collected research confirmed that graphene-coated textile electrodes are machine washable and maintain signal quality comparable to conventional Ag/AgCl electrodes ³. This validates statements about **washability and durability**. However, we must ensure that such claims are framed as general properties of carbon or graphene electrodes used in textile wearables, rather than implying a specific collaboration with ETH Zurich.

2 Device Return Process

The comments clarified that SKIIN kits include a **prepaid return label** rather than a prepaid envelope. Users should place the used kit back in its original box, affix the prepaid return label provided inside the

box, and drop it at a postal location. This is a minor yet important correction to the copy to prevent confusion.

3 Clinician Workflow: MVCP vs. Holter Analysis Software

The current specification references a single physician portal but does not distinguish between the **MVCP (Medical Vital Cloud Portal)** and the **Holter Software Analysis solution**. Research from internal documentation notes the following distinctions:

- **MVCP (Medical Vital Cloud Portal):** An administrative platform that allows healthcare providers to **set up Holter studies**, manage patients, and review ongoing ECG data during the monitoring period. GPs can see raw ECG traces and annotated symptom events via MVCP. This platform supports real-time oversight and administrative tasks.
- **Holter Software Analysis (Medical Algorithmics platform):** After the monitoring period is complete, the collected data is processed by AI and reviewed by cardiac technicians using the Holter analysis software. The AI pre-processing flags potential arrhythmias; cardiac technicians and cardiologists then review the data and **generate the final diagnostic report**. This final report is what is sent to the prescribing physician and patient.

The current spec lumps these under a single “Physician portal” bullet. To improve clarity and accuracy for clinicians, the updated specification should explicitly describe both components: **MVCP** for setting up and monitoring studies, and the **Holter analysis platform** for AI-assisted review and report generation.

4 Gaps & Misstatements Identified

1. **Incorrect attribution of sensor development:** The specification currently claims the garment’s sensors were “Swiss-engineered in collaboration with ETH Zurich textile labs.” Evidence suggests the technology originates from Myant’s textile computing R&D and Nanoleq’s patents ¹. The final copy should credit Myant and note the integration of Nanoleq’s stretchable cables and dry electrodes.
2. **Return process description:** The spec instructs users to “Use the prepaid envelope to send the kit back,” but the actual process involves a **prepaid return label**. The copy must reflect this to ensure users return the kit correctly.
3. **Clinician workflow clarity:** The distinction between MVCP and Holter analysis software is missing. Clinicians need to know that MVCP is for study setup and real-time review, while the Holter analysis platform is used by technicians and cardiologists to generate the final report. This nuance is important for transparency and accuracy.
4. **Evidence for compliance claims:** The spec still refers to “Swiss-engineered” in a way that might mislead about ETH Zurich involvement. Additionally, claims like “95 % compliance rate” should be avoided unless validated by published studies. Our updated clinical evidence section already softened this to “high patient compliance observed in studies,” which is acceptable.

5 Research Takeaways for Final Spec

- **Update sensor attribution:** Replace any mentions of “Swiss-engineered sensors” or “collaboration with ETH Zurich” with a description of **textile computing technology developed by Myant**,

incorporating **Nanoleq's patented dry electrodes and stretchable conductive yarns** ¹ .

Emphasise the integration of monitoring capabilities through knitting and conductive textiles.

- **Correct return instructions:** Specify that users should **repackage the kit, apply the prepaid return label found in the box**, and drop it in the mail. Avoid mentioning envelopes.
- **Clarify clinician journey:** Break out two separate sub-sections: **MVCP** (for set-up and real-time monitoring) and **Holter Analysis** (AI pre-processing plus technician review to generate the final report). Explain how clinicians interact with each platform.
- **Maintain general claims backed by research:** Continue to highlight washability, comfort, high compliance, and extended monitoring benefits as previously validated, ensuring all statements remain evidence-based.

By addressing these gaps and integrating the new information, the final copy specification will be more accurate, transparent, and aligned with the actual technology and workflow of SKIIN.

¹ ² Myant acquires Swiss smart textile companies

<https://www.innovationintextiles.com/myant-acquires-swiss-smart-textile-companies/>

³ Carbon-Based Textile Sensors for Physiological-Signal Monitoring - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10253857/>